

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	6 September 2026
Team ID	Individual
Name	Noshma Chowdary Uppuluri
Project Name	Employee Raise Issue – Record Producer & Service Portal Integration
Maximum Marks	4 Marks

Technical Architecture:

The proposed solution uses ServiceNow as the application platform. Employees submit workplace or technical issues through the Employee Center using a Record Producer. UI Policies and dependent choices ensure structured and accurate information, while the submitted issue is stored in a custom ServiceNow table. Support users can manage the issue, assign it to the appropriate group, update its status and priority, and track its progress through the ServiceNow platform..

Architecture Flow

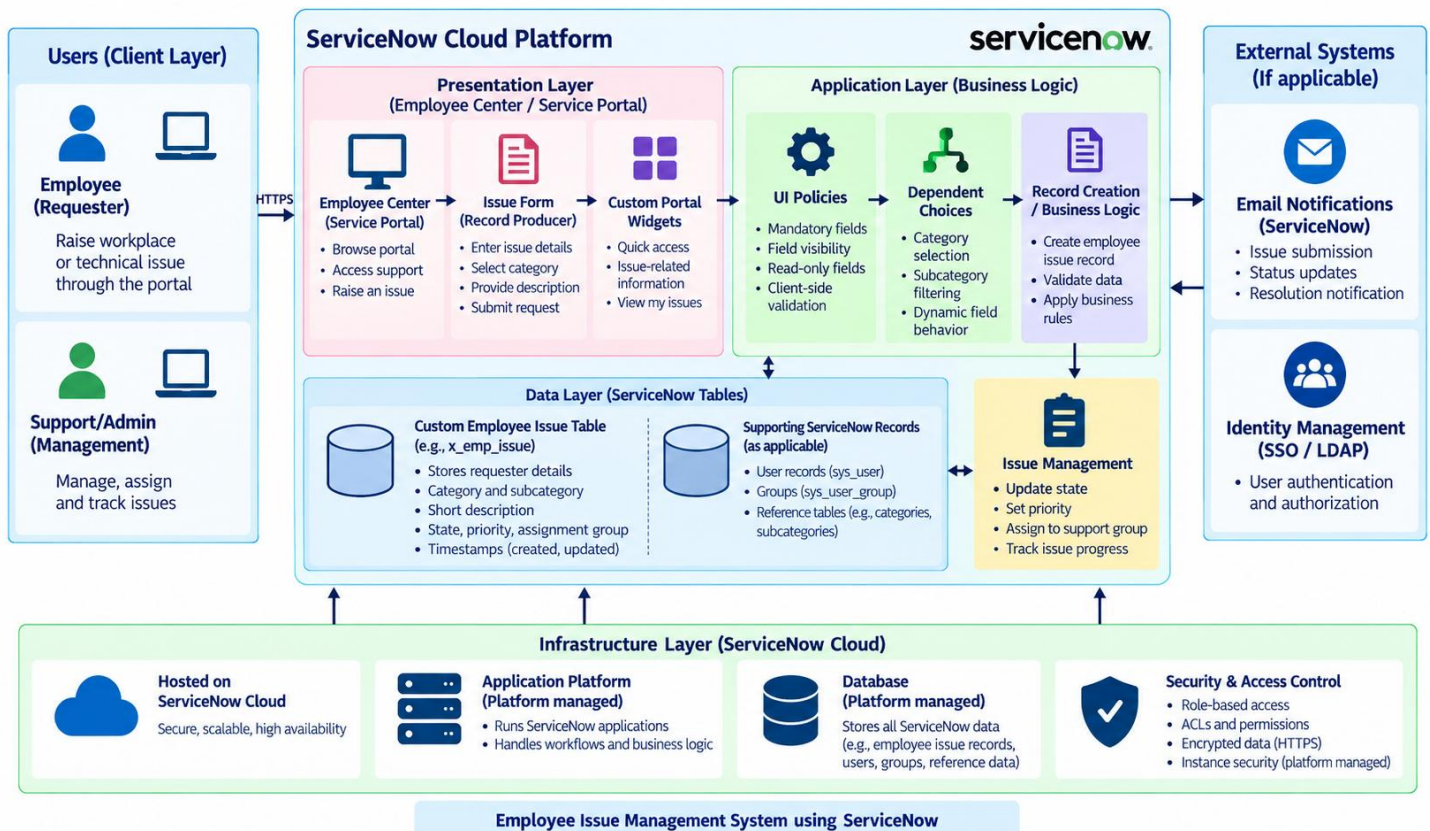


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Interface through which employees access the Employee Center and submit workplace or technical issues.	ServiceNow Employee Center / Service Portal
2	Application Logic-1	Captures employee issue details and creates the corresponding issue record.	Record Producer, ServiceNow Forms & Variables
3	Application Logic-2	Controls field behavior and validation while the employee submits an issue.	ServiceNow UI Policies
4	Application Logic-3	controls category and subcategory selection and displays relevant choices based on the selected category.	ServiceNow Dependent Choices / Choice Lists
5	Database	Stores employee issue information including requester, category, subcategory, description, state, priority and assignment details.	Custom ServiceNow Table / ServiceNow Tables
6	Cloud Database	Provides platform-managed storage for the application's issue records and supporting ServiceNow data.	ServiceNow Platform Data Store
7	File Storage	No separate file-storage component is required for the core issue-reporting workflow.	Not applicable
8	External API-1	No external API is required for software request processing.	Not applicable
9	External API-2	No external API is required for software request processing.	Not applicable
10	Machine Learning Model	No machine-learning model is required for the proposed workflow.	Not applicable
11	Infrastructure (Server / Cloud)	The application, portal and business logic are hosted and executed on the ServiceNow cloud platform.	ServiceNow Personal Developer Instance / Enterprise Instance

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No separate open-source application framework is required. The solution is configured within the ServiceNow platform.	ServiceNow platform configuration
2	Security	Access to employee issue records and administrative functions is controlled using ServiceNow authentication, roles, permissions and access controls.	ServiceNow roles, ACLs, authentication and authorization
3	Scalable Architecture	The solution is platform-based and can support additional software catalog items, users, approval rules, and fulfillment tasks as the request volume grows.	ServiceNow platform and Flow Designer
4	Availability	The application runs on the ServiceNow platform, allowing employees and support teams to access the configured issue-management system through the ServiceNow instance.	ServiceNow instance/Employee Center
5	Performance	The solution reduces manual issue-reporting effort by providing a structured form, controlled data entry, automated record creation and organized issue tracking.	Record Producer, UI Policies & ServiceNow Tables